

Catenaries, Curve Fitting and the Three Forms of a Quadratic

Chain & Curve Lab · Custom (Claude AI)

Year 11	~30 minutes	Mathematics · Algebra & Functions
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NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: [simulations/chain-curve-lab.html](https://simulations.chain-curve-lab.html) — open it from the STEM Simulation Hub

Curriculum links: QCAA Mathematical Methods (General) · Unit 1, Topic 2 — Functions and Graphs (turning points, axes of symmetry, intercepts · the discriminant · transformations of $f(x)$ · the vertical line test · verifying and evaluating a model)

Learning intentions

- I can determine the turning point, axis of symmetry and x-intercepts of a quadratic from its graph, and connect these features to the standard, turning-point and intercept forms of its equation.
- I can solve a quadratic equation by completing the square and using the quadratic formula, and use the discriminant to explain how many real solutions it has.
- I can use the vertical line test to explain why a relation is or isn't a function, and describe how the parameters of a function transform (translate or dilate) its graph.

Getting started

Open the Chain & Curve Lab simulation. It has three modes, switched at the top: **Discover** (a physically simulated hanging chain), **Challenge** (guess the function yourself), and **Forms** (build an exact parabola from three points). This worksheet uses Forms mode for Tasks 1–5, and Discover mode for Tasks 6–7.

Tasks

1. Switch to **Forms** mode and click **Reset points** so $P1(-2, 0)$, $P2(1, 0)$, $P3(0, -2)$. Record the **Standard form** shown. Substitute $x = -2$ by hand to confirm $y = 0$, showing $P1$ really does lie on this curve.

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2. Using the standard form from Task 1, complete the square by hand to derive the **turning-point form**. Show every step, then check your answer against the simulation's own turning-point form.

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3. Use the quadratic formula on the standard form from Task 1 to find the x-intercepts by hand. Compare your two solutions to the simulation's **intercept form** and to the discriminant shown in the Features panel — what does the sign of the discriminant tell you about how many real solutions exist?

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4. Complete this table using the Features panel in the simulation.

Feature	Value shown in the simulation
Vertex	
Axis of symmetry	
y-intercept	
x-intercept(s)	

5. Drag P2 (or edit its coordinates) so it shares an x-coordinate with P1 but has a different y-coordinate. What message does the simulation show? Using the **vertical line test**, explain why no function can pass through two points that share an x-coordinate but not a y-coordinate.

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6. Switch to **Discover** mode and let the chain settle. Its equation has the form $y = a \cdot \cosh((x-h)/a) + k$ — the same a , h , k roles as the general transformation $y = a \cdot f((x-h)/a) + k$. Record a , h and k , then drag the right peg sideways only (same height) and describe which parameter changes and why, in terms of translating the graph horizontally.

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7. Tick both **Catenary** and **Parabola** under Fitted models. Compare the two models' RMSE at Slack length = 5% and again at Slack length = 100%. Using your results, evaluate the usefulness of a parabola as a model for a hanging chain — qualitatively (in words) and quantitatively (using the RMSE numbers).

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Extension (optional)

A ball's height h (metres) is recorded at three times: $t = 0$ s, $h = 1$ m; $t = 1$ s, $h = 21$ m; $t = 2$ s, $h = 21$ m. Using the same method as Forms mode — solving three simultaneous equations — find $h(t) = at^2 + bt + c$ by hand. State the ball's maximum height and when it occurs.

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